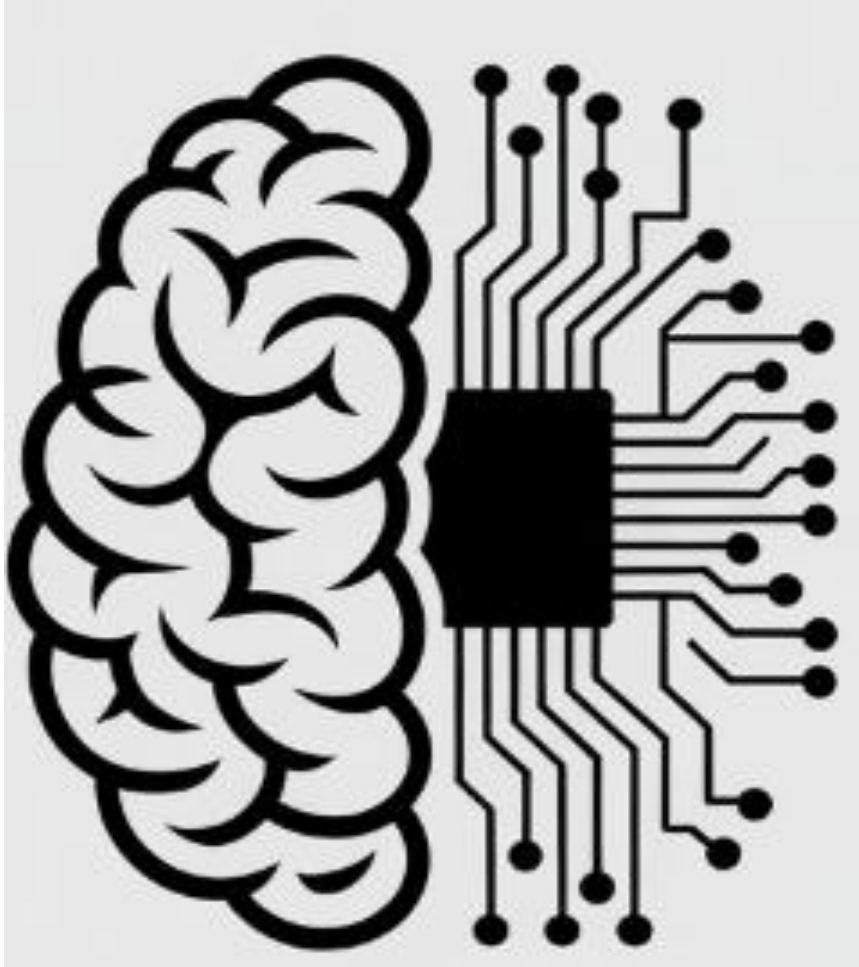




Neural Network Issues

Dr. Sobia Tariq Javed

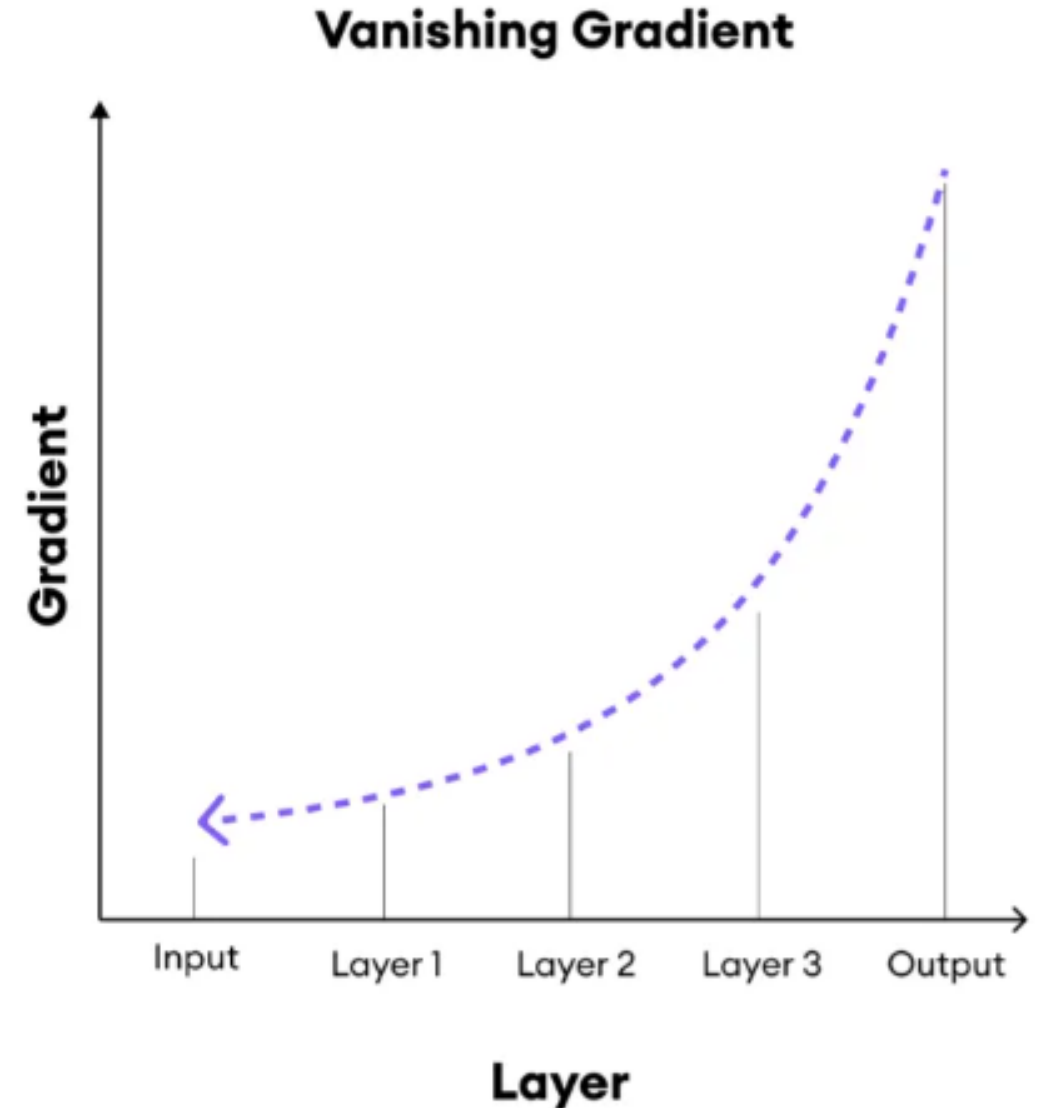
Challenges in Training Neural Network

- **1) Vanishing Gradient**

- It is a problem where **gradients** become very small during **backpropagation**, preventing **deep neural networks** from learning effectively.
- During **backpropagation**, gradients are multiplied across many layers.
- If **gradients** are very **small** (less than 1), they keep **shrinking**.
- Early layers receive **near-zero** updates.
- The network **stops learning** effectively.

Challenges in Training Neural Network

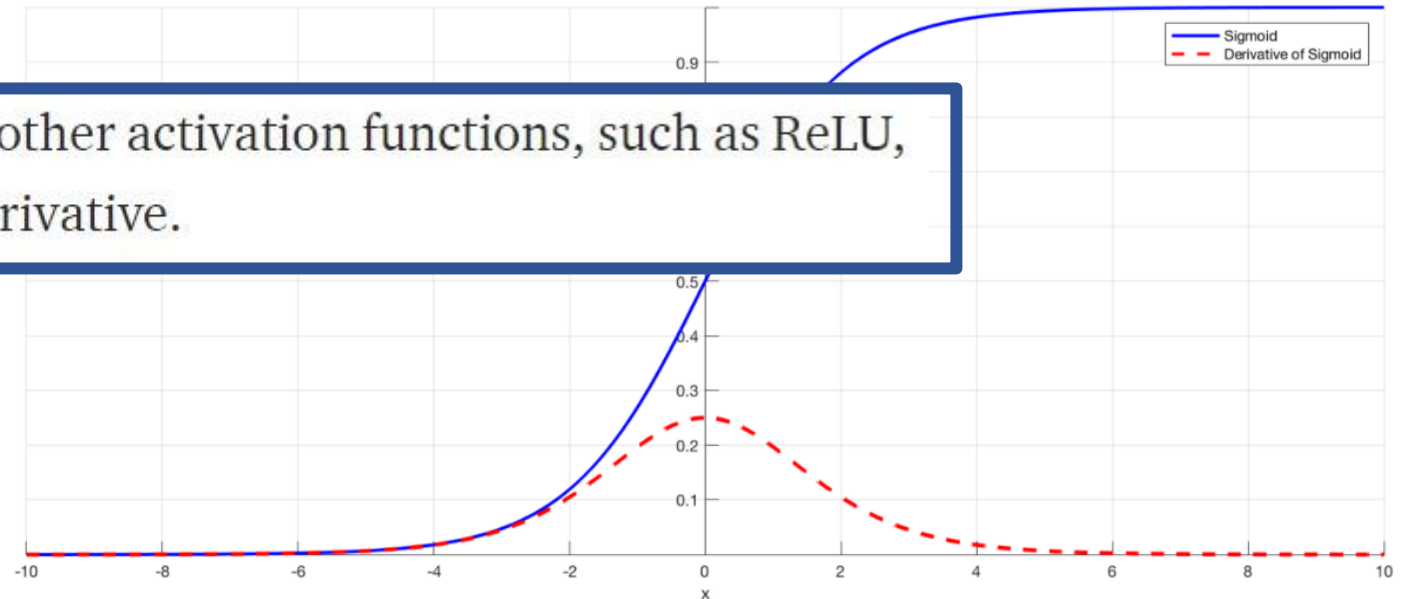
- 1) **Vanishing Gradient- Solution**
 - ReLU activation
 - Batch normalization
 - Proper weight initialization



Challenges in Training Neural Network

- The gradients of the loss function **approaches zero**, making the network hard to train.

The simplest solution is to use other activation functions, such as ReLU, which doesn't cause a small derivative.



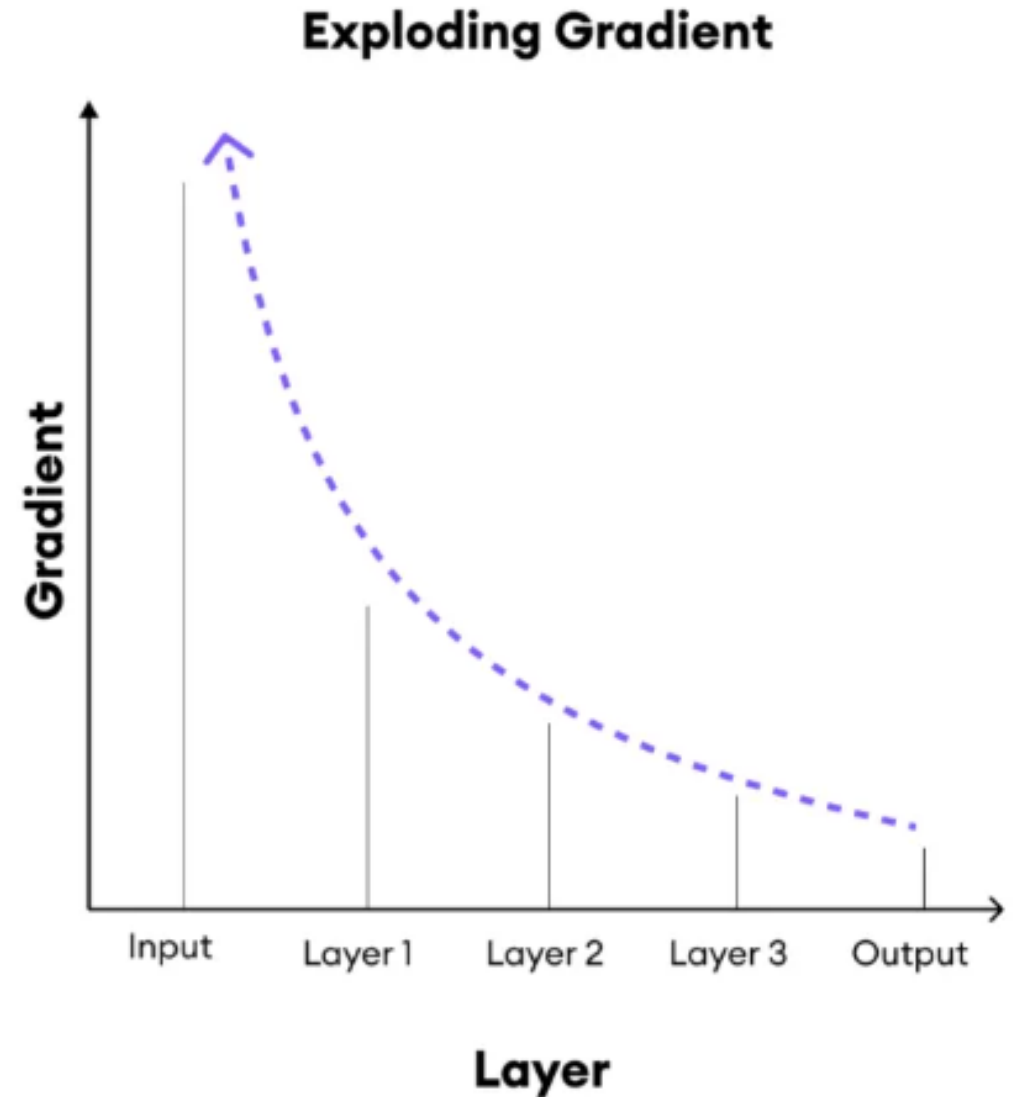
Challenges in Training Neural Network

- 2) **Exploding Gradient**

- It is a problem where **gradients** grow too large during **back propagation**, causing unstable learning and large weight updates.
- **Large gradients** cause very **large weight updates** during training.
- **Weights** may become **unstable** and **grow uncontrollably**.
- The model may produce **NaN values** or **infinite loss**.
- **Training** becomes **unstable** and the network fails to **converge**.
- This problem commonly occurs in **deep neural networks**.

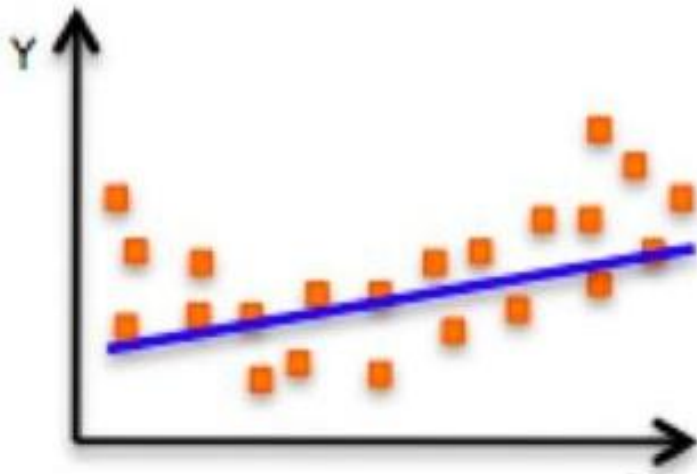
Challenges in Training Neural Network

- 2) **Exploding Gradient** - Solution
 - Gradient clipping
 - Proper weight initialization
 - Batch normalization
 - Smaller learning rate



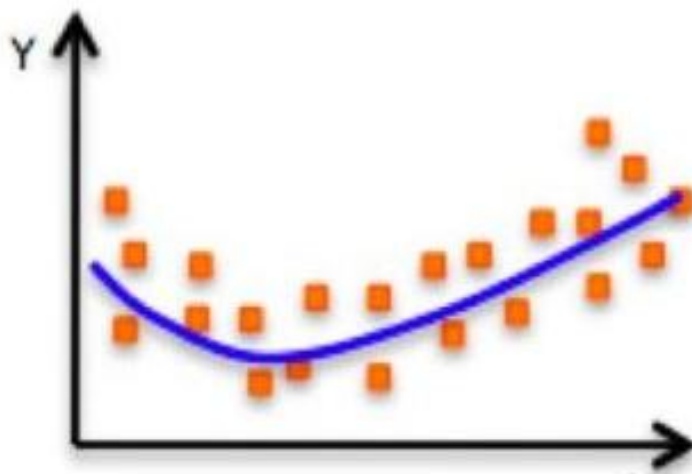
Overfitting and Underfitting

- These describe how well a model generalizes from training data to unseen data.

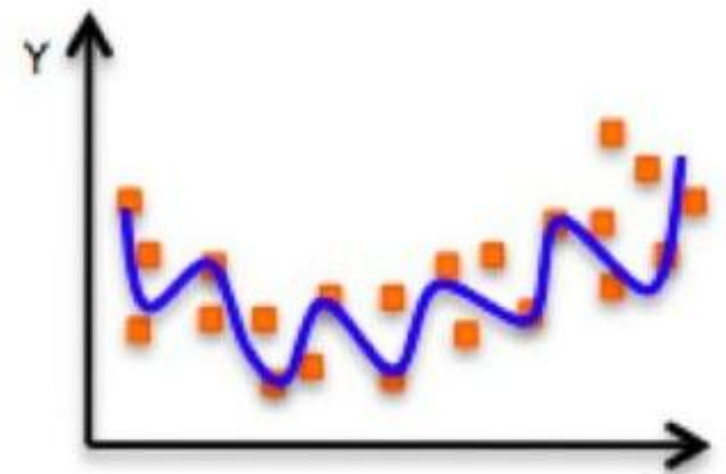


Underfitting

Model does not have capacity to fully learn the data



Ideal fit

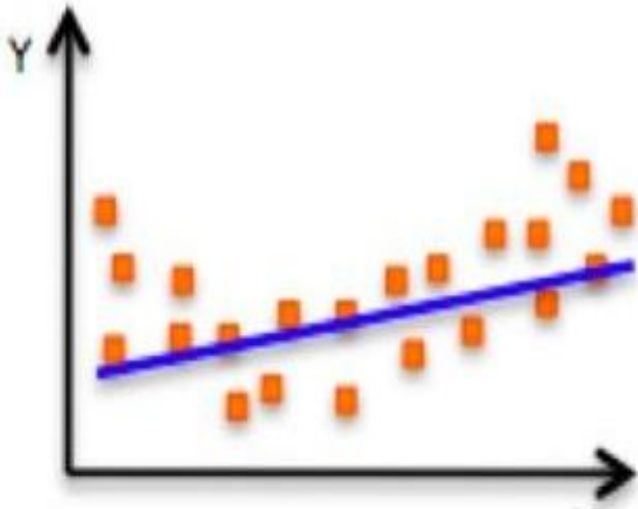


Overfitting

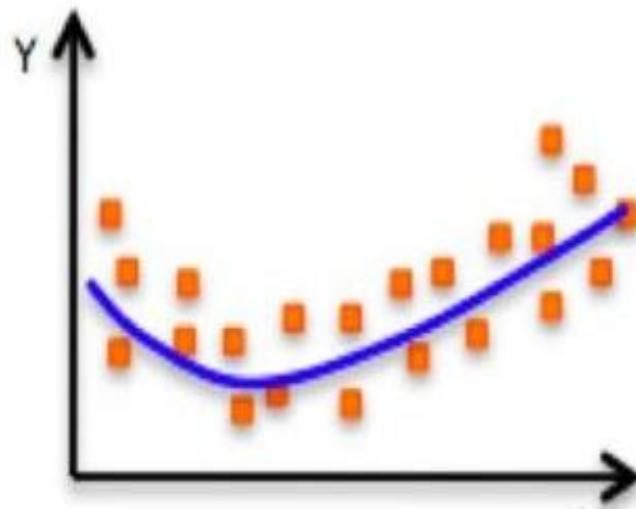
Too complex, extra parameters, does not generalize well

Underfitting

- It occurs when the **model is too simple** to capture the underlying pattern in the data.
- The model fails to learn even the **training data** properly.



Underfitting



Balanced

- **Characteristics**
 - Training error --- 
 - Testing error --- 
 - **Poor performance** everywhere

Underfitting

- **Bias**

- It is the error caused by **oversimplified assumptions** in the model.
- It measures how far the **model's predictions** are from the **true relationship**.

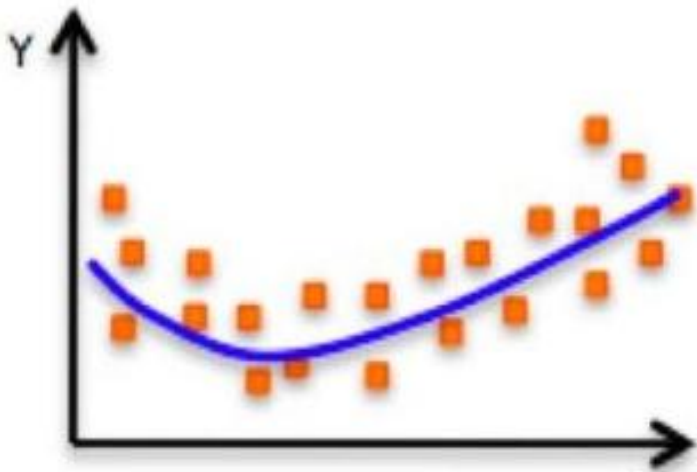
Solution:

- Increase **model complexity**
- Train longer (**more epochs**)
- Reduce **regularization**
- Add better **features**

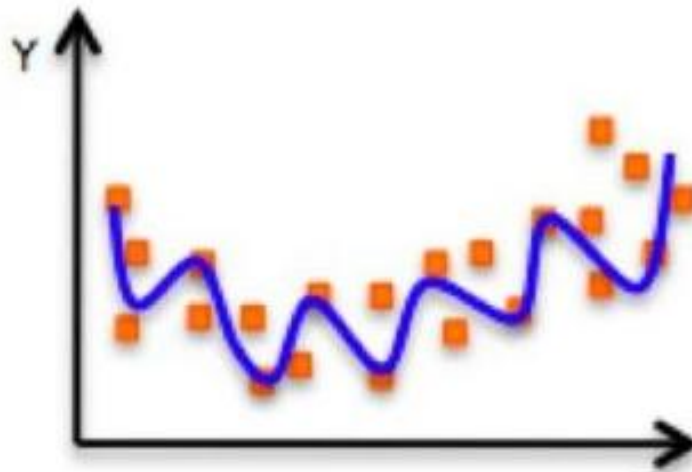
High Bias = Model
is too simple

Overfitting

- Model memorizes training data instead of learning general patterns, leading to poor performance on new data.



Balanced

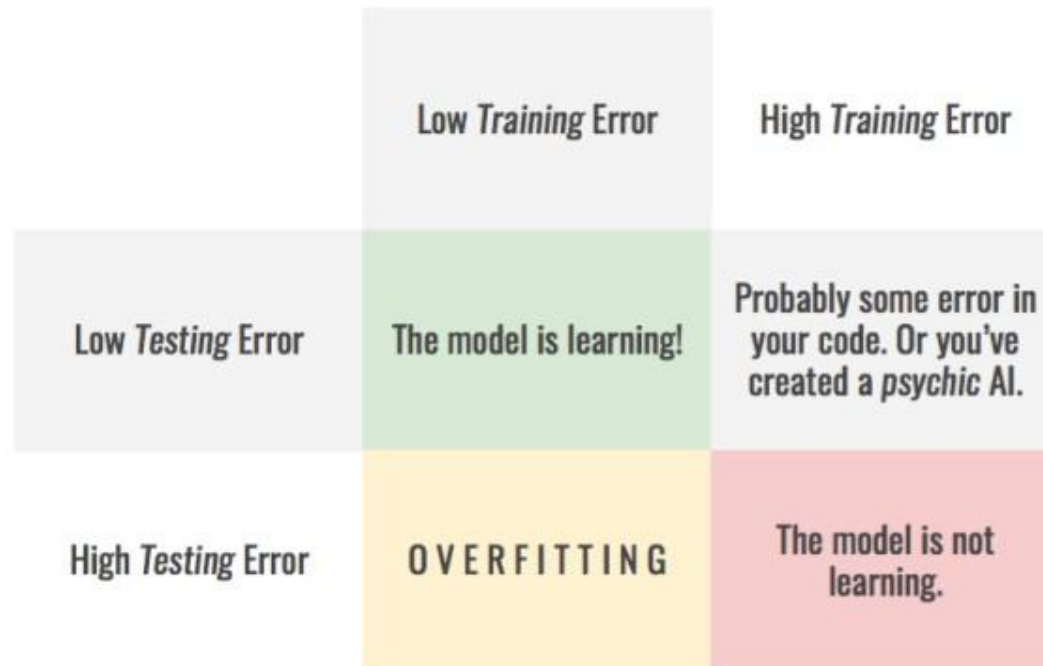


Overfitting

- **Characteristics**
 - Training error --- ↓
 - Testing error --- ↑
 - **Poor Generalization**

Overfitting

- To test this ability, a simple method consists in splitting the dataset into two parts: the training set and the test set.



Overfitting can be seen as the **difference** between the **training** and **testing** error.

Overfitting

- **Variance**

- It is the error caused by **model sensitivity** to small changes in **training data**.
- It measures how much **predictions change** if the **training data changes**.

High Variance =
Model is too
sensitive (overfits)

Solution:

- Collect more **Data**
- Simplify the **model**
- **Regularization**
- **Dropout** Regularization
- **Data Augmentation** and **noise**
- **Early stopping**

Overfitting

- Validation Vs. testing Data
- N-cross validation

Bias Vs. Variance

Case	Bias	Variance	Result
Underfitting	High	Low	Poor learning
Good Fit	Balanced	Balanced	Good generalization
Overfitting	Low	High	Poor generalization

Bias – Variance Tradeoff

- Cannot minimize both simultaneously — improving one often worsens the other.
- Increase model complexity:
 - Bias ↓
 - Variance ↑
- Decrease complexity:
 - Bias ↑
 - Variance ↓

Simplify the Model



On the left, the model is too simple. On the right it overfits.

Reference

- Chapter 4 of Tom M. Mitchell “Machine Learning”